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The need for built-in validation of surveillance data so that changes in diagnostic performance of post-mortem meat inspection can be detected

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Abstract

Time trends in animal-disease surveillance often are evaluated on the basis of crude estimates of apparent prevalence. In addition to possible changes in the true prevalence of the condition, changes in apparent prevalence over time might reflect changes in sensitivity and/or specificity of the diagnostic classification used. To illustrate this, comparative post-mortem meat inspection data from four Danish slaughter plants sampled in 1993–1994 and 1997–1998 were used to obtain latent-class model estimates of the sensitivity and specificity of traditional and extended post-mortem meat inspection of visceral and parietal chronic pleuritis (CP), respectively.

True prevalence of CP was estimated for each study period and slaughter plant by latent-class models. Estimated sensitivities of traditional post-mortem meat (TPM) inspection ranged from 28.8 to 61.4% (1993–1994) and 39.2 to 87.3% (1997–1998). An increase in sensitivity with time was seen for all slaughter plants. Estimated sensitivities of extended post-mortem meat (EPM) inspection ranged from 85.7 to 94.8% (1993–1994) and 73.8 to 93.0% (1997–1998). All estimated specificities were >93.3%.

The possible association of the estimated true prevalence of CP with time (1993–1994 versus 1997–1998) was investigated with a logistic-regression model with random effects. A slight, but non-significant decrease in the odds of CP from 1994 to 1998 was found (odds ratio = 0.9).

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In this and similar situations, one should consider conducting ongoing double-classification of samples of units followed by statistical estimation of true prevalences, sensitivities and specificities, so that decisions can be based on such estimates rather than on crude apparent prevalences.

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1. Introduction

Time trends in animal-disease surveillance commonly are evaluated on the basis of crude estimates of apparent prevalence, as seen in relation to the Danish *Salmonella enterica* surveillance and control program in pigs (Mousing et al., 1997), the Danish *Streptococcus agalactiae* surveillance program in cattle (Andersen and Huda, 1995), the European bovine spongiform encephalopathy (BSE) surveillance and control program (Anon., 2000) and the traditional post-mortem meat (TPM) inspection statistics from slaughtered livestock. Chronic pleuritis (CP) is a frequent finding of the TPM inspection of Danish slaughter pigs. The apparent prevalence of CP has increased steadily since the beginning of the 1980s (Vraa-Andersen, 1994); e.g. the proportion of pigs with this lesion was 0.14 in 1987, increasing to 0.25 of the total kill in 2000 (Christensen, G., 2001, personal communication). However, it is quite possible that changes in apparent prevalence over time reflect changes in sensitivity (Se) and/or specificity (Sp) of the test used—not just changes in the true prevalence. The potential fallacy of making inferences based on apparent rather than true prevalence may be considerable, especially when non-detection rates are high as with post-mortem meat inspection data.

To assess the possible impact (on estimates of prevalence) associated with changes in the Se and Sp of a test over time, we used comparative post-mortem meat inspection data from four Danish slaughter plants. Extended post-mortem meat (EPM) inspection (Christensen et al., 1999) was carried out on a sample of plucks from slaughter pigs and compared to data from the TPM inspection in 1993–1994 (Christensen, 1995); a similar study was performed in 1997–1998 (Christensen and Enøe, 1999).

Our objectives were to assess the validity of TPM and EPM inspection procedures for the detection of CP and to investigate whether the estimated true prevalence of CP in 1997–1998 compared with the level in 1993–1994.

2. Materials and methods

2.1. Populations under study

The study was conducted at four Danish slaughter plants: A, B, C and D situated in the northern, middle and southern part of the peninsula of Jutland and in the island of Zealand, respectively. The populations under study comprised all pigs that were slaughtered at these four plants in the last quarter of 1993 and the first three quarters of 1994 (Christensen, 1995) and similarly in the last quarter of 1997 and the first three quarters of 1998 (Christensen and Enøe, 1999). The combined total annual kill at these four plants was

6,146,354 and 6,110,782 in the study periods in 1993–1994 and 1997–1998, respectively. This amounts to approximately 32% of the total annual kills of about 19 million pigs in 1993–1994 and in 1997–1998.

2.2. Sampling

The slaughter plants were chosen because they were situated in four different geographical areas of Denmark. The investigation comprised comparative inspections of 8860 plucks and carcasses in 1993–1994 and 10,102 in 1997–1998. The size of the samples was determined out of practical and economic considerations. The sampling was done quarterly in both study periods. The total sample consisted of $4 \times 4 \times 2 = 32$ sub-samples—each consisting of approximately 600 plucks/carcasses. The samplings were carried out in December, March, June and September. The date of sampling was selected non-randomly. On the day of sampling, approximately every second carcass on the slaughter line was examined by EPM inspection. Carcasses were sampled from the same slaughter line in all four samplings within each study period. The total number of kills at the four plants on the days of sampling were 78,272 (1993–1994) and 100,050 (1997–1998). The samples made up 11.3% of the total number of kills at the four plants on the days of sampling in 1993–1994 and 10.1% in 1997–1998, respectively. An additional sample of 728 carcasses was evaluated at plant B in March 1999 to investigate the agreement between TPM and EPM inspection when carried out by the same inspector.

The visceral pleura of the lungs and the parietal pleura of the chest wall were examined for presence of CP by EPM and TPM inspection (Christensen et al., 1999). While the TPM inspection is an overall assessment of pathological findings in the carcass and organs, the EPM inspection is targeted on specific groups of organs. In this instance, the EPM inspection provided data on pathological changes in the lungs and pleura that were far more detailed than data gathered during the TPM inspection. A skilled inspector performed the EPM inspection of all samples and in both study periods applying a standardized procedure (Christensen et al., 1999). Data from the EPM inspection were entered into a mobile terminal on-site and subsequently merged with TPM inspection data from the slaughter plant database. Data were supplemented with information from the Danish SPF organization (Finn Kildam, personal communication, 1999), which provided information about the health status of the herds concerned. The herds were classified into three groups based on their health status: specific pathogen-free (SPF) herds, MS herds and conventional herds. SPF herds in Denmark are declared free of *Actinobacillus pleuropneumoniae* serotype 2, *Mycoplasma hyopneumoniae*, toxin-producing *Pasteurella multocida*, *Brachyspira hyodysenteriae*, *Sarcoptes scabiei* and *Haematopinus suis*. Certain serotypes of *A. pleuropneumoniae* may be accepted within the SPF program (such as serotype 6). SPF herds infected with *M. hyopneumoniae* are designated as MS herds.

2.3. Statistical methods

Latent-class models implemented by Gibbs sampling (Enøe et al., 2000) were used to obtain estimates of true prevalences of CP and of the Se and Sp of TPM and EPM inspection (with their 95% Bayesian confidence intervals (CIs)) for each year and each

slaughter plant. The Bayesian approach was preferred to the frequentist method because of the ease with which the Bayesian 95% CIs were derived. Non-informative prior distributions were used for the unknown parameters because high quality a priori information was unavailable. The latent-class model requires two or more sub-samples for the estimation, each allowing cross tabulations of the number of findings of CP as determined by each of the two diagnostic procedures in question (Hui and Walter, 1980). In this study, three sub-samples were identified for each year and plant by the health status of the supplying herds (SPF, MS and conventional) and prevalence estimates were obtained for each of the three sub-samples. The two diagnostic methods were assumed to be conditionally independent, i.e. that the outcome of one test is independent of the outcome of the other test, given the true disease state. The assumption of conditional independence was confirmed as described by Enøe et al. (2001).

For each year and slaughter plant, the three health-specific prevalence estimates were weighted to a prevalence estimate using weights according to the number of observations in each sub-sample. The Gibbs sampler was used to obtain the posterior distribution of the health-specific estimates and of the weighted estimates and subsequently Bayesian 95% CIs were calculated.

Apparent prevalences of CP based on EPM and TPM inspection data were calculated with their 95% CI. TPM and EPM inspections of the additional 728 carcasses collected at plant B were carried out by the same experienced inspector. First the visceral pleura was inspected and pathological changes recorded and subsequently this was repeated for the parietal pleura in a similar way. The inspections were not blinded or carried out independently of each other. Kappa was used to determine the degree of agreement between the two diagnostic methods.

The possible association of the prevalence of CP with time was investigated with a logistic-regression model with random effects. The analysis was performed using samples of 8860 pigs (1993–1994) and 10,102 pigs (1997–1998) from a total of 2206 herds with sufficient information. The herd-specific animal-level true prevalence (p_{herd}) was modeled as:

$$\log \text{it}(p_{\text{herd}}) = \text{year} + \text{health} + \text{season} + \text{plant} + \beta \text{ pigs} + \text{herd}$$

where the explanatory variables were year (1993–1994 versus 1997–1998), health status (health) of the herd (SPF, MS, conventional), season (quarters 1–4), slaughter plant (plant) (A–D) and number of pigs (pigs) supplied per herd on the days of sampling (as a proxy for herd size). Herd was a random effect of herd, $\text{herd} \sim N(0, \sigma^2)$, which described the herd-to-herd variation not covered by the covariates (i.e. the overdispersion). Different transformations were used to investigate the linearity assumption for the continuous variable pigs.

The observations from a herd were n_{11} , n_{12} , n_{21} , n_{22} , where n_{11} was the number of animals with a positive finding of CP in both diagnostic procedures, $n_{12}(n_{21})$ were the number of CP-positive animals detected only by extended (traditional) post-mortem meat inspection and n_{22} was the number of animals tested negative in both. The observations (n_{11} , n_{12} , n_{21} , n_{22}) followed a multinomial distribution (N , p_{11} , p_{12} , p_{21} , p_{22}), where:

$$N = n_{11} + n_{12} + n_{21} + n_{22}$$

$$p_{11} = p_{\text{herd}} \text{Se}_1 \text{Se}_2 + (1 - p_{\text{herd}})(1 - \text{Sp}_1)(1 - \text{Sp}_2)$$

$$p_{12} = p_{\text{herd}} \text{Se}_1(1 - \text{Se}_2) + (1 - p_{\text{herd}})(1 - \text{Sp}_1)\text{Sp}_2$$

$$p_{21} = p_{\text{herd}}(1 - \text{Se}_1)\text{Se}_2 + (1 - p_{\text{herd}})\text{Sp}_1(1 - \text{Sp}_2)$$

$$p_{22} = p_{\text{herd}}(1 - \text{Se}_1)(1 - \text{Se}_2) + (1 - p_{\text{herd}})\text{Sp}_1\text{Sp}_2$$

The Se and Sp of EPM (Se_1 and Sp_1) and TPM (Se_2 and Sp_2) inspection were regarded as known and equal to the estimates from the Gibbs sampler (Table 3), i.e. slaughter plant and year specific.

3. Results

The observed distribution of CP as determined by TPM and EPM inspection stratified according to health status of the supplying herds in 1993–1994 and 1997–1998 is shown in Table 1.

Table 1

The observed distribution of CP as determined by TPM and EPM inspection (+: presence; -: absence) stratified according to health status of the supplying herds and year (18,962 pigs processed in 1993–1994 and 1997–1998 at four Danish slaughter plants)

Plant	Health status	TPM+		TPM−	
		EPM+	EPM−	EPM+	EPM−
1993–1994					
A	SPF	11	3	4	39
	MS	32	12	18	220
	Conventional	441	92	306	1071
B	SPF	1	0	6	125
	MS	14	1	47	175
	Conventional	216	28	566	1055
C	SPF	0	1	3	20
	MS	42	25	35	264
	Conventional	410	93	338	1046
D	SPF	5	1	15	131
	MS	9	0	25	213
	Conventional	204	40	324	1133
1997–1998					
A	SPF	13	7	2	41
	MS	103	34	19	297
	Conventional	528	241	120	1239
B	SPF	3	0	6	74
	MS	52	3	77	242
	Conventional	316	35	562	1286
C	SPF	6	1	13	172
	MS	107	21	100	392
	Conventional	356	96	294	1083
D	SPF	5	2	6	118
	MS	14	7	8	110
	Conventional	317	122	146	1306

Table 2

Estimated true weighted ($\hat{P}_w$; %) and apparent prevalence (AP; %) determined by EPM and TPM inspection of parietal and visceral CP (18,962 pigs processed in 1993–1994 and 1997–1998 at four Danish slaughter plants)

Plant	Years	Number in sample	$\hat{P}_w^a$	CI ^b	AP _{EPM}	CI ^c	AP _{TPM}	CI ^c	AP _{TPM} total kill
A	1993–1994	2249	38.7	33.9, 43.3	36.1	34.1, 38.1	26.3	24.5, 28.1	26.0
	1997–1998	2644	37.2	29.2, 42.3	29.7	28.0, 31.4	35.0	33.2, 36.8	34.8
B	1993–1994	2234	39.7	34.7, 43.6	38.1	36.1, 40.1	11.6	10.3, 12.9	11.7
	1997–1998	2656	38.1	31.9, 42.3	38.3	36.5, 40.1	15.4	14.0, 16.8	14.5
C	1993–1994	2277	35.3	29.9, 40.6	36.4	34.4, 38.4	25.1	23.3, 26.9	24.8
	1997–1998	2641	36.5	31.3, 40.7	33.2	31.4, 35.0	22.2	20.6, 23.8	22.9
D	1993–1994	2100	26.7	21.8, 31.8	27.7	25.8, 29.6	12.3	10.9, 13.7	17.7
	1997–1998	2161	26.2	20.7, 30.9	23.0	21.2, 24.8	21.6	19.9, 23.3	23.4

^a Weighted according to the number of observations in each of the three health-specific sub-samples.

^b 95% Bayesian CIs by Gibbs sampling.

^c 95% CIs.

Considerable variation in the apparent prevalence of parietal CP, as determined by TPM inspection was seen between slaughter plants and years (Table 2). An increase by inspection in apparent prevalence from 1994 to 1998 was observed at slaughter plants A, B and D. Apparent prevalences of parietal CP were generally lower than estimated true weighted prevalences (Table 2). Apparent prevalences of parietal CP in the samples were close to the apparent prevalences based on all kills in the study periods except for slaughter plant D where sample estimates were lower especially in 1994 (Table 2).

Estimated true weighted prevalences ranged from 26.2 to 39.7% (Table 2) and they varied only slightly between slaughter plants. However, the true prevalence at slaughter plant D appeared to be lower than at the other plants. Apparent prevalences of visceral CP and estimated true weighted prevalences were fairly similar (Table 2).

Estimated Ses of TPM inspection ranged from 28.8 to 61.4% (1993–1994) and 39.2 to 87.3% (1997–1998) (Table 3). An increase in Se with time was seen for all slaughter plants (Table 3).

Estimated Ses of EPM inspection ranged from 85.7 to 94.8% (1993–1994) and 73.8 to 93.0% (1997–1998) (Table 3). A decrease in Se was seen for slaughter plants A, C and D and a slight increase was seen for B (Table 3). All estimated Sps were >93.3% (Table 3). Bayesian 95% CIs for Se and Sp estimates are shown in Table 3.

The observed distribution of CP in the sample of 728 pigs examined by the same rater was: 387 pigs were CP-positive by both diagnostic procedures, 28 (44) were CP-positive detected only by extended (traditional) post-mortem meat inspection and 269 were found CP negative by both. A comparison showed some disagreement in the outcomes of the two diagnostic methods ($\kappa = 0.8$).

There was no significant association of the estimated true prevalence of CP with time (1993–1994 versus 1997–1998) in the logistic-regression model (Table 4). A slight decrease in the odds of CP from 1994 to 1998 was found (odds ratio = 0.9). However, there was a significant effect of slaughter plant—the odds of CP were lowest in slaughter

Table 3

Estimated sensitivity (Se; %) and specificity (Sp; %) of TPM and EPM inspection for detection of parietal and visceral CP (18,962 pigs processed in 1993–1994 and 1997–1998 at four Danish slaughter plants)

Plant	Year	TPM inspection				EPM inspection			
		Se	CI ^a	Sp	CI ^a	Se	CI ^a	Sp	CI ^a
A	1993–1994	61.4	57.3, 67.8	95.9	92.6, 99.3	90.2	82.4, 98.8	98.3	93.2, 99.9
	1997–1998	87.3	81.2, 98.5	96.7	87.9, 99.9	73.8	68.0, 90.1	96.9	92.1, 99.9
B	1993–1994	28.8	25.2, 33.0	99.6	98.1, 99.9	90.6	85.6, 97.6	96.6	91.1, 99.8
	1997–1998	39.2	35.0, 46.5	99.3	97.6, 99.9	93.0	88.7, 99.2	95.5	88.1, 99.8
C	1993–1994	59.0	53.3, 67.3	93.3	91.0, 96.4	94.8	85.9, 99.7	95.2	89.2, 99.7
	1997–1998	59.8	53.6, 68.1	99.4	97.2, 99.9	81.2	77.3, 86.7	94.4	89.5, 98.6
D	1993–1994	45.2	37.6, 55.2	99.6	98.3, 99.9	85.7	80.3, 92.4	93.3	88.6, 98.7
	1997–1998	77.1	67.3, 91.8	98.4	94.3, 99.9	75.9	69.7, 88.9	96.1	92.0, 99.5

^a 95% Bayesian CIs by Gibbs sampling.

Table 4

Associations between CP and year (1993–1994 versus 1997–1998) in 18,962 Danish slaughter pigs (multivariable logistic-regression, adjusted for effect of slaughter plant (A–D), season of year, health status of supplying herds, number of supplied pigs in study period and herd effect)

Explanatory variable	<i>b</i>	S.E. (<i>b</i>)	<i>P</i>	OR ^a	95% CI (OR)
Year (1993/1994 versus 1997/1998)	−0.11	0.09	<0.20	0.9	0.8, 1.1
Quarter (March versus December)	0.18	0.10	0.07	0.8	0.7, 0.9
Quarter (June versus December)	0.34	0.10	0.008	1.4	1.2, 1.7
Quarter (September versus December)	0.12	0.11	0.27	1.1	0.9, 1.4
Slaughter plant (A versus D)	0.61	0.13	<0.001	1.9	1.4, 2.4
Slaughter plant (B versus D)	0.66	0.12	<0.001	1.9	1.5, 2.5
Slaughter plant (C versus D)	0.57	0.13	<0.001	1.8	1.4, 2.3
Health status (MS versus SPF)	1.67	0.31	<0.001	5.3	2.9, 9.8
Health status (conventional versus SPF)	2.41	0.29	<0.001	11.1	6.3, 19.6
Number of supplied pigs	4.4×10^{-3}	9.3×10^{-4}	<0.001	—	—

^a For an increase in number of supplied pigs by 10, the OR = 1.04.

plant D and highest in slaughter plants A and B. Furthermore, there was an effect of season—the odds of CP were lowest in pigs slaughtered in March and highest in pigs slaughtered in June. There were significant effects of health status (the highest odds in pigs from conventional and the lowest in pigs from SPF herds) and number of supplied pigs (odds increases with number of pigs). In a reduced model in which health status and number of slaughter pigs were excluded, the effect of year remained the same.

4. Discussion

Crude estimates of apparent prevalence of parietal CP were highly biased when compared to estimated true weighted prevalences—leading to considerable underreporting of parietal CP especially in 1994 (but also to a lesser extent in 1998).

The apparent increase in prevalence of CP from 1994 to 1998 in 3 of 4 slaughter plants might be explained by an increase in the Se of the TPM inspection. Intensified slaughter processes leave less time to remove CP from the chest wall, so that the meat inspectors have to assign the lesion code of CP to the carcasses—leading to further processing on a separate slaughter line and a reduced non-detection rate. Moreover, results from the investigation in 1994 (Christensen, 1995) indicated that TPM inspection had a fairly low Se; the report might have led to an increased awareness of the inspectors and thereby an improved Se of TPM inspection.

The apparent prevalences of CP determined by TPM inspection based on data from the samples and on the total kills (Table 2) were very alike, except for plant D, especially in 1993–1994. An explanation for this could be that the pigs in the two study periods were sampled from different lines of slaughter at plant D and that there could be some variation of the Se of the post-mortem meat inspection between different lines of slaughter.

The Bayesian estimates of Se and Sp showed a fairly large variation between slaughter plants—indicating differences in the criteria used for classifying CP.

A comparison of the methods by paired observation in 728 pigs showed that despite the belief that parietal and visceral pleuritis are manifestations of the same disease condition, it was not possible to obtain perfect agreement. If the pathological changes are not measures of exactly the same underlying latent condition, there might be a slight bias in the latent-class estimates of Se, Sp and true prevalence.

Because the slaughter plants contributing to the present study process approximately 1/3 of all pigs slaughtered in Denmark, the conclusions might apply to the whole population.

Results from the logistic-regression analyses substantiated that there was no increase in true prevalence of CP, when data from 1993 to 1994 and 1997 to 1998 were compared. The logistic-regression estimates of the effect of year did not change when health status and number of supplied pigs were excluded from the models—indicating that changes over time in production patterns and management could not account for the changes in apparent prevalences.

Furthermore, these results indicate that post-mortem meat inspection data—as well as other surveillance data—should be validated carefully before they are used in national disease statistics or research projects. In this and similar situations, one should consider conducting ongoing double-classification of samples of units followed by statistical estimation of true prevalences, Ses and Sps, so that decisions can be based on such estimates rather than on crude apparent prevalences. This methodology is of course only possible if two or more diagnostic tests or methods are available and applicable. An advantage of the latent-class methodology is that even if the diagnostic methods are far from being perfect, they still can be used in the surveillance and will yield valid estimates of the true prevalence. The methodology is especially useful in the surveillance of fairly common conditions and might be applied in the Danish *Salmonella* surveillance program. However, we caution that when the methodology is applied to post-mortem meat inspection data, there seems to be some variation between slaughter plants; estimates of Se, Sp and prevalence therefore are slaughter plant specific. It is not possible at present to apply the latent-class methodology when the condition of interest is assumed to be very rare (such as with the BSE-surveillance program). This is because prevalences in the

populations will almost be zero (and very close) and as a consequence, estimation is not possible (Hui and Walter, 1980).

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